Stochastic Programming

Chapter 4: Information value and Stochastic solution

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Contents

Outline

1. The Expected Value of Perfect Information
2. The Value of Stochastic Solution
3. Basic Inequalities

Definition

- The "Expected Value of Perfect Information" (*EVPI*) measures the maximum amount a decision maker would be ready to pay in return for complete information about the future.

- The usual stochastic problem is:

$$\begin{array}{ll} \min & z(x) = c^\top x + E_\xi \left[\begin{array}{ll} \min & q^\top y \\ \text{s. to.} & Wy = h - Tx \\ & y \geq 0 \end{array} \right] \\ \text{s. to.} & Ax = b \\ & \qquad\quad x \geq 0 \end{array}$$

- q, h and T may depend on ω, i.e. ξ=[$q(\omega)$, $h(\omega)$, $T(\omega)$].

- Remember $K_1 = \{x : Ax = b, x \geq 0\}$

WS Solution

Let's consider the following problem:

$$\begin{aligned}\min_{x\in K_1} z(x,\xi) = c^\top x + \min\ & q(\xi)^\top y \\ \text{s. to. } & Wy = h(\xi) - T(\xi)x \\ & y \geq 0\end{aligned}$$

- This is a deterministic optimization problem, with solution denoted by $\overline{x}(\xi)$ and optimal value $z(\overline{x}(\xi), \xi)$ (if they exist).
- If, somehow, we knew in advance the value of the r.v. ξ, we could get the ideal solution of the stochastic problem, known as the "Wait-and-See" solution (WS). The value of this solution would be:

$$\boxed{WS = E_\xi[\min_{x\in K_1} z(x,\xi)] = E_\xi[z(\bar{x}(\xi),\xi)]}$$

Example

Consider the SP:

$$\begin{array}{ll} \min x + & E_\xi \left[\begin{array}{ll} \min & 2y \\ \text{s. to.} & y \geq \xi - x \\ & y \geq 0 \end{array} \right] \\ \text{s. to.} & \\ & x \geq 0 \end{array} \qquad \text{where } \xi \sim U[0,1]$$

For each scenario represented by each ξ, we formulate the particular problem:

$$\begin{array}{ll} \min & x + 2y \\ \text{s. to.} & y \geq \xi - x \\ & x, y \geq 0 \end{array} \rightarrow \quad \begin{array}{ll} \min & x + 2y \\ \text{s. to.} & y + x \geq \xi \\ & x, y \geq 0 \end{array} \rightarrow \quad \begin{array}{l} \bar{x}(\xi) = \xi \\ \bar{y}(\xi) = 0 \\ z(\bar{x}(\xi), \xi) = \xi \end{array}$$

Therefore,

$$WS = E_\xi[\min z(x,\xi)] = E_\xi(\xi) = \frac{1}{2}$$

EVPI

Consider now the solution of the original stochastic problem, written in terms of $z(x, \xi)$:

$$\boxed{RP = \min_{x \in K_1} E_\xi[z(x,\xi)]}$$

From now on this problem will be called the "Recourse Problem" (RP), and we will assume it has an optimal solution x^*:

- Clearly, this is the stochastic problem:

$$\min_{x \in K_1} E_\xi \left[\begin{array}{lll} c^\top x + & \min & q^\top y \\ & \text{s. to.} & Wy = h - Tx \\ & & y \geq 0 \end{array} \right] = \min_{x \in K_1} c^\top x + E_\xi \left[\begin{array}{ll} \min & q^\top y \\ \text{s. to.} & Wy = h - Tx \\ & y \geq 0 \end{array} \right]$$

- The Expected Value of Perfect Information is defined as

$$\boxed{EVPI = RP - WS}$$

Cont. Example

We compute *RP*

$$RP = \min_{x \geq 0} x + E_\xi \left[\begin{array}{ll} \min & 2y \\ \text{s. to.} & y \geq \xi - x \\ & y \geq 0 \end{array} \right]$$

Since $\xi \sim U[0,1]$, optimal solution is

$$y = \max\{0, \xi - x\} = \left\{ \begin{array}{ll} \xi \geq x \rightarrow & y = \xi - x \\ \xi < x \rightarrow & y = 0 \end{array} \right.$$

Thus, $\mathcal{Q}(x) = E_\xi(Q(x,\xi)) = \displaystyle\int_0^1 2\max\{0, \xi - x\} f(\xi) d\xi = \int_0^x 0 \cdot 1 d\xi + \int_x^1 2(\xi - x) \cdot 1 d\xi =$ $\displaystyle\int_x^1 2\xi d\xi - 2x \int_x^1 d\xi = \xi^2|_x^1 - 2x(1-x) = x^2 - 2x + 1 = (x-1)^2$

Then, $RP = \min_{x \geq 0} x + \mathcal{Q}(x) = \min_{x \geq 0} x + (x-1)^2$. Deriving, we obtain $x^* = \frac{1}{2}$, and finally

$$RP = \frac{1}{2} + \left(\frac{1}{2} - 1\right)^2 = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$EVPI = RP - WS = 3/4 - 1/2 = 1/4$$

Outline

Definitions

- If compute the Recourse Problem *RP* is difficult, and the Wait-and-See problem *WS* could be impossible, we may compute the value of the Expected Value Solution *EV*.

$$\boxed{EV = \min_{x \in K_1} z(x, \bar{\xi})}$$

Solution $\bar{x}(\bar{\xi})$ of the *EVP*, replacing the ξ occurrences by $\bar{\xi} = E(\xi)$.

- In general, this value makes a little insecurity as first stage decision. If we do and put the solution in our *RP*, fixing the first stage decisions $\bar{x} = \bar{x}(\bar{\xi})$, we get the "Expected result of using the *EV* solution":

$$\boxed{EEV = E_{\xi}[z(\bar{x}(\bar{\xi}), \xi)]}$$

This measures how good decision $\bar{x}(\bar{\xi})$ performs.

- The Value of Stochastic Solution *VSS* is defined as follows:

$$\boxed{VSS = EEV - RP}$$

This quantity is the cost of ignoring uncertainty in choosing a decision.

Example

- Compute EV, EEV, VSS in last example.

$$RP = \min_{x \geq 0} x + E_\xi \left[\begin{array}{ll} \min & 2y \\ \text{s. to.} & y \geq \xi - x \\ & y \geq 0 \end{array} \right]$$

Replace $\xi \sim U[0,1]$ by $E(\xi) = 1/2$, and solve RP.

We can solve the stochastic problem as well as the recourse problem:

$$\begin{array}{ll} \min & x + 2y \\ \text{s. to.} & y + x \geq 1/2 \\ & x, y \geq 0 \end{array}$$

$$\Downarrow$$

$$\left\{ \begin{array}{l} x^* = 1/2 \\ y^* = 0 \end{array} \right.$$

$$\begin{array}{ll} \min & 2y \\ \text{s. to.} & y \geq 1/2 - x \\ & y \geq 0 \end{array}$$

$$\Downarrow$$

$y^* = \max\{1/2 - x, 0\}$ and then:

$$\min x + 2\max\{1/2 - x, 0\} \Rightarrow \left\{ \begin{array}{l} x^* = 1/2 \\ y^* = 0 \end{array} \right.$$

Now, he have $\bar{x}(1/2) = 1/2$, $\bar{y}(1/2) = 0$ and $EV = z(\bar{x}(1/2), 1/2) = 1/2$.

Example Cont.

- Compute EEV:

$$EEV = E_\xi[z(\bar{x}(1/2), \xi)] = \min 1/2 + \quad E_\xi \left[\begin{array}{ll} \min & 2y \\ \text{s. to.} & y \geq \xi - 1/2 \\ & y \geq 0 \end{array} \right]$$

$$EEV = 1/2 + E_\xi[2\max\{0, \xi - 1/2\}] = 1/2 + (1/2 - 1)^2 = 1/2 + 1/4$$

Finally, $VSS = EEV - RP = 1/2 + 1/4 - (1/2 + 1/4) = 0$

- In this example, there is no benefit for using the *RP* solution instead the solution of *EEV*.

EVPI & VSS ROADMAP

$$\begin{array}{lll} \min & z(x) = c^\top x + E_\xi \left[\begin{array}{ll} \min & q^\top y \\ \text{s. to.} & Wy = h - Tx \\ & y \geq 0 \end{array} \right] \\ \text{s. to.} & Ax = b \\ & x \geq 0 \end{array} \equiv RP = \min_{x \in K_1} E_\xi[z(x,\xi)] = z(x^*)$$

RP vs. best possible (ideal) solution

$$z(\bar{x}(\xi), \xi) = \min_{x \in K_1} z(x,\xi) = c^\top x + \begin{array}{ll} \min & q^\top y \\ \text{s. to.} & Wy = h - Tx \\ & y \geq 0 \end{array}$$

$(q(\xi), h(\xi), T(\xi))$

Optimal x w.r.t. ξ

$Q(x, \xi)$

$$WS = E_\xi[\min_{x \in K_1} z(x,\xi)] = E_\xi[z(\bar{x}(\xi),\xi)]$$

$$EVPI = RP - WS$$

RP vs. deterministic sol. ($\bar{\xi}$)

$\xi := \bar{\xi}$

$$EV = \min_{x \in K_1} z(x, \bar{\xi})$$

$\bar{x}(\bar{\xi})$

Solve RP with $x := \bar{x}(\bar{\xi})$

$$EEV = E_\xi[z(\bar{x}(\bar{\xi}),\xi)] = z\left(\bar{x}(\bar{\xi})\right)$$

$$VSS = EEV - RP$$

Outline

Propositions

Some basic inequalities are used to obtain fast bounds of *VSS* and *EVPI* values. This lets us know if it's worth RP calculated or how far we stay from *WS*.

Proposition 1

$$\boxed{WS \leq RP \leq EEV}$$

Proposition 2

For a Stochastic Problem with fixed q, and W:

$$\boxed{EV \leq WS}$$

Propositions

Proposition 3

- For any Stochastic Program, it is verified that

$$EVPI \geq 0$$

$$VSS \geq 0$$

- For a Stochastic Program with fixed q and W :

$$EVPI \leq EEV - EV$$

$$VSS \leq EEV - EV$$

Conclusion

- These useful results tell us that
  - $EVPI \geq 0 \rightarrow$ Knowing the perfect information is never negative.
  - $VSS \geq 0 \rightarrow$ Using the stochastic solution is never worse than using the expected value solution.
- We can obtain some limits of *EVPI* and *VSS* computing *EV* and *EEV* (expected value solution, and expected result of using this value) , which are relatively easy to compute. If $EEV - EV$ is small or big, we may wonder:
  - whether is worth computing a solution of the stochastic problem (which is more expensive).
  - whether is worth invest to know more about future

When $EEV - EV$ is small, probably the answer of last questions is NO.

Example revisited

In the example we had:

$$EEV = \frac{1}{2} + \frac{1}{4} \qquad EV = \frac{1}{2} \qquad WS = \frac{1}{2} \qquad RP = \frac{1}{2} + \frac{1}{4}$$

- $EEV - RP = VSS \leq EEV - EV \qquad \longleftarrow 0 \leq \frac{1}{4}$
- $RP - WS = EVPI \leq EEV - EV = \frac{1}{4} \qquad \longleftarrow \frac{1}{4} \leq \frac{1}{4}$